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ENG EK301 A4, Fall 2025

Final Design Report

December 5, 2025

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Introduction:

The final truss structure was designed to maximize the supported load while adhering to a set of constraints and minimizing the cost while taking into account restrictions. The most challenging constraint was the length restriction, which immediately disqualified many unique designs. The analysis program found that using the shorter member lengths in a triangle formation positioned close to the load joint created the strongest designs. To lower cost we eliminated zero force members and gave our design the lowest profile possible. After taking this information into account we found that our truss design materialized quickly as there was little room for creativity with so many considerations.

Procedure:

The primary change in our design procedure was in the process of drawing truss designs. In our preliminary design we would draw them out on paper, creating the coordinates first, and then drawing members in between. We discovered that using a computer-aided design (CAD) software such as OnShape or Autodesk Fusion 360 was significantly more effective in design. These programs allowed us to shift away from coordinate based design and into truss designs based on member length. Member length based designs are superior to coordinate based ones because the strength of the truss lies in the member lengths, not the placement of joints. With these programs, member lengths could be set and the overall shape could be dragged by leaving certain member lengths free. The programs allowed for measurement of individual member lengths as well as the truss' total length. Therefore the use of CAD allowed us to streamline the design process and improve the strength of our final truss.

Analysis:

The analysis of our designs remained similar to our preliminary method and mostly improved based on the quantity of designs evaluated. In the graphing section of the Matlab program we added critical member highlighting and load joint identification which allowed us to better visualize the failure of the truss. Our initial design had a zero force member as well as an oversized profile, creating extra length. In an effort to eliminate these faults our design had strengths of low material and an efficient size.

Most of the improvements in our analysis were from plugging in significantly more designs into our Matlab code and understanding their performance. We found that asymmetrical trusses with the load focused to one side performed better, and trusses with more minimum length members were successful.

The uncertainty for the T value was acquired from the buckling lab curve fit. The uncertainty in the $w_{Failure}$, was calculated using the standard formula for error propagation:

$$W_{failure} = P_{crit} * W_{applied} / T_c$$

$$\sigma P_{crit} = \pm 1.58\sigma z$$

No uncertainty in $W_{applied}$ and therefore no uncertainty in T_c makes $W_{applied} / T_c$ a constant. The formula for propagation of uncertainty in a constant multiple $B = kA$ is:

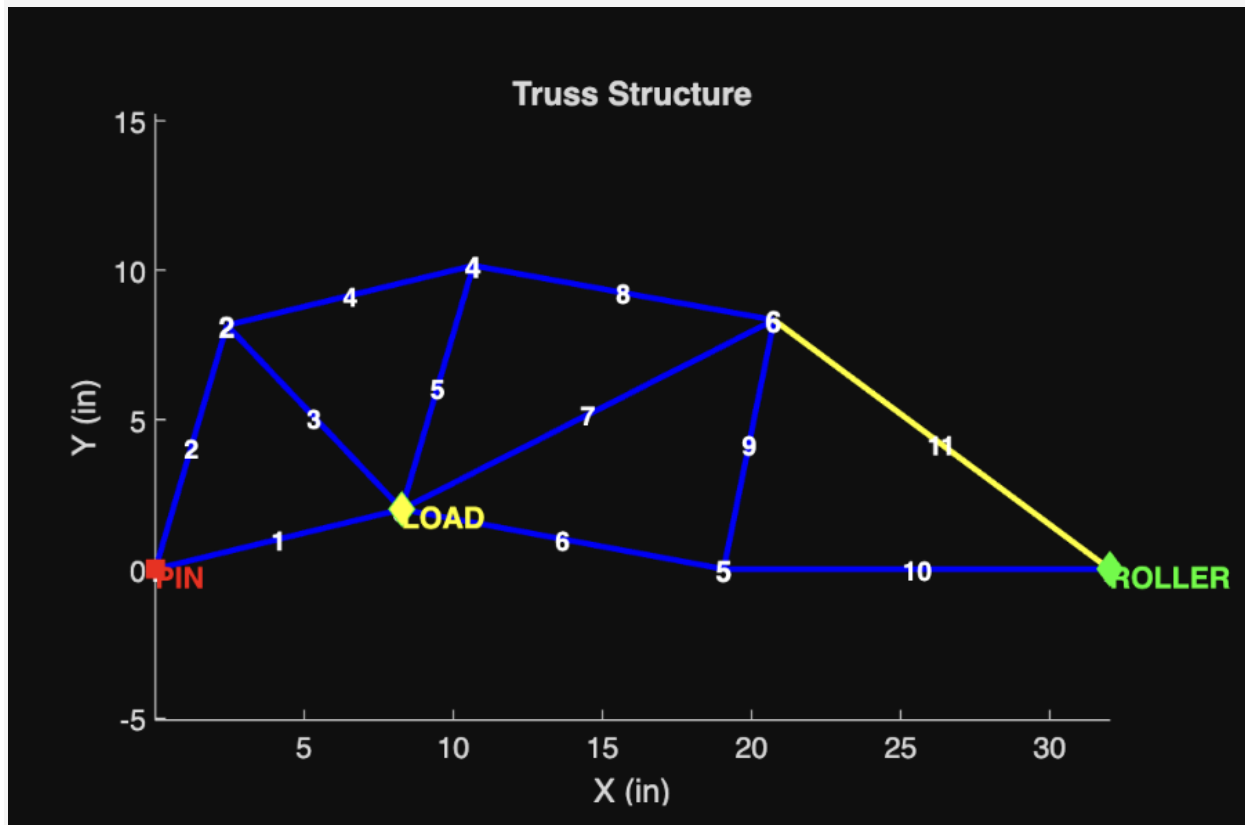
$$\sigma B = k\sigma A$$

Therefore:

$$\sigma W_{failure} = (W_{applied} / T_c) * \sigma P_{crit}$$

Results:

The final design:



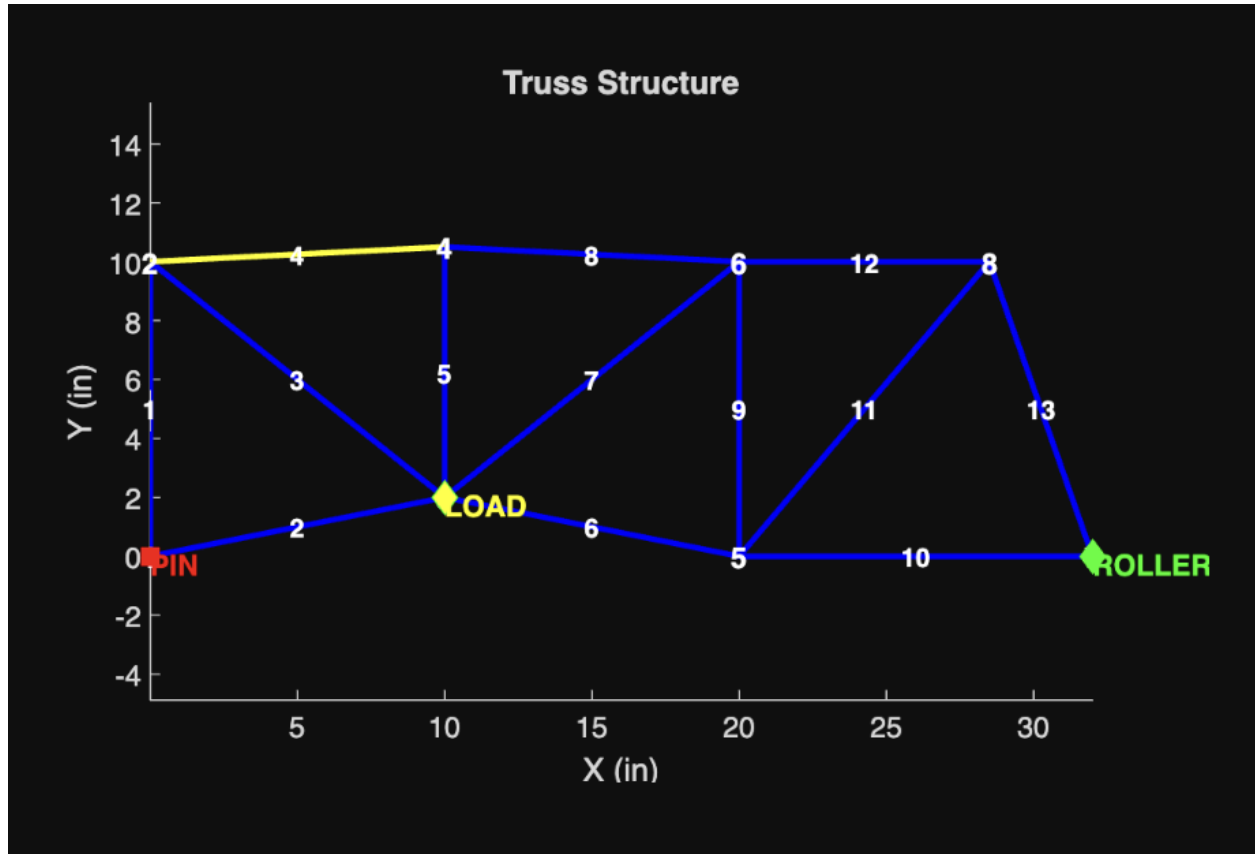
Member:	Force Experienced (oz)	Tension or Compression	Member Length (in.)	Maximum Buckling Force (oz)
m1	7.735	Tension	8.5	
m2	26.641	Compression	8.5	46.1083 +/- 1.58
m3	26.641	Tension	8.5	
m4	26.641	Compression	8.5	46.1083 +/- 1.58
m5	10.838	Tension	8.5	

m6	10.937	Tension	10.981	
m7	13.545	Tension	14.0	
m8	23.205	Compression	10.2519	33.2791 +/- 1.58
m9	2.032	Compression	8.5	46.1083 +/- 1.58
m10	11.158	Tension	12.9402	
m11	13.884	Compression	14.0	19.3511 +/- 1.58

Total Cost:	\$183.17
Load/Cost Ratio	.243 oz/\$
Total Length	113.1731 in
Critical Member	Member 11
Load Joint	Joint 3
Maximum Load	44.602 +/- 3.642oz

Discussion & Conclusion:

The original design that we based on final design off of was this box truss:



This design was significantly over the maximum length allowed for this project, however, it was quite cheap to manufacture at 0.196 oz./\$. It also had a good failure weight of 42.818 ± 1.951 oz, highlighting its potential as a candidate for further development. With some improvements, such as shortening some members and removing certain sections to simplify the overall design, we reached the final truss shown previously. The most useful aspect of this project was certainly the development of the MATLAB program. If we could change or revise anything about this project, we would reduce the certain parameters (such as minimum member length) to encourage more creativity in the designs.

Appendix

Hartford Civic Center Area Collapse - Meeting Minutes

Date and time: 12/4/25 5:20 PM

Location: Boston University Ingalls Engineering Study Lounge, 44 Cummington Mall, Boston, MA 02215

Group Members: Mukund Balaji, Silas Lockwood, Mark Travník

Chair: Mark Travník

Recorder: Mukund Balaji

Planned Agenda:

- 1) Review and summarize the Hartford Civic Center roof collapse.
- 2) Discuss the role of computer programs in engineering analysis and design.
- 3) Identify failures (technical, procedural, ethical).
- 4) Determine an appropriate safety factor for our truss design.
- 5) Discuss realistic maximum load to claim for our truss, considering variability in construction and materials.
- 6) Reiterate relevance of NSPE Code of Ethics, Fundamental Canon #1.

Meeting Minutes:

- 1) Summary
 - a) Mukund summarized the event: The Hartford Civic Center Arena roof collapsed in 1978 due to design deficiencies and snow load exceeding the roof's actual weakened capacity. Although the venue was at full capacity only a few hours earlier, there were no fatalities.

- b) Silas mentioned: During construction there were signs of deflection beyond the expected amount but it was dismissed.

2) Use of Computer Programs in the Design

- a) Mark stated that the engineers relied heavily on computer analysis for this complex design.
- b) Mukund added that the computer program used did not properly account for the complex forms buckling that the frame would experience.
- c) Silas said that computers aren't wrong, they can only compute with the assumptions they've been given.
- d) Mark stated that the overconfidence in computer systems created an environment where engineers didn't question the calculated strength of the truss.

3) Technical Failures

- a) Mark listed several key design flaws from the report:
 - i) The dead load was underestimated by about 20%.
 - ii) Missing bracing on compression members (actual bracing every 30 ft instead of 15 ft).
 - iii) Unusual member sections that were built with low buckling resistance.
 - iv) Horizontal and diagonal bars did not intersect at the same node.
- b) Mukund mentioned that some of the top layers were overstressed 850%, showing that the base design was extremely unstable.

4) Procedural Failures

- a) Mark brought to attention that the passing of responsibility between many contractors led to many errors.

- b) Silas talked about how the construction manager refused to hire an independent structural engineer for inspection.
 - c) Mukund added that many deflections after the structure was erected were completely ignored.
 - d) Silas mentioned that the city did not require peer review for such a unique design.
- 5) Ethical Analysis (NSPE Canon #1)
- a) Mukund stated the 1st Canon: "Engineers shall hold paramount the safety, health, and welfare of the public."
 - b) Mark claimed that the fact this structure failed, meant that this rule was violated
 - c) Silas noted that public projects should never solely rely only on theoretical computer calculations.
 - d) Mukund states the importance of conducting safety checks, especially when theoretical data does not match with real-world data.
 - e) Silas mentioned that failing to notice or call to attention early signs of failure in the truss also violate the 1st Canon.
- 6) How to incorporate these lessons on our truss
- a) If our truss were to be used in the real world, where public safety would be a concern, we would design the truss such that it has a large factor of safety, which depends on the materials used to build the truss.
 - b) Real world factors like wind, misalignment, and bending can impact the structural integrity of the truss, even if it may not be present in a computer program.
 - c) During the construction of a truss, early warning signs of failure should not be disregarded.

- d) With this factor of safety in mind, a good maximum load (in real life) should be around one-third of the theoretical maximum load.

Conclusions:

- During truss construction make sure to keep the fabrication as similar as possible to the assumptions made in the Matlab program.
- Keep the maximum load above $\frac{4}{3}$ of the desired minimum. Therefore the load must be at least $\frac{4}{3} \cdot 32 = 42.667$ oz
- Watch for unexpected buckling when constructing the truss.
- Be sure to point out any potential issues during truss construction, not worrying about ego or feelings but prioritizing the strength of our design.
- Manual verification of the model as well as constant skepticism in its reliability is the best atmosphere for avoiding overreliance in the program.